

A Universal Framework for Persistence Between Nullity and Closure

Primitive Structure

Let a system be a triple

$$\mathcal{S} := (X, F, O)$$

where:

- X is a nonempty state space,
- $F : X \rightarrow X$ is an evolution operator,
- $O : X \rightarrow Y$ is an observation map.

Define the trajectory of an initial state $x_0 \in X$ as

$$x_t := F^t(x_0), \quad t \in \mathbb{N}.$$

Axioms

Axiom 1 (Observational Nullity). *A state $x \in X$ is null if there exists an equivalence class $[x] \subseteq X$ such that*

$$\forall y \in [x], \quad O(y) = O(x).$$

Axiom 2 (Reference). *A system admits reference if there exists a function*

$$R : X \rightarrow \{0, 1\}$$

such that for some $x, y \in X$, $R(x) \neq R(y)$.

Axiom 3 (Persistence). *A system persists if there exists $x_0 \in X$ such that:*

$$\forall t \in \mathbb{N}, \quad F^t(x_0) \text{ is defined}$$

and

$$\forall N \in \mathbb{N}, \exists t > N \text{ such that } O(x_t) \neq O(x_{t-1}).$$

Axiom 4 (Closure Idealization). *A system is closed if there exists a total internal procedure deciding all properties of its own future evolution.*

Definitions

Definition 1 (Lower Bound WI).

$$\text{WI} := \{x \in X \mid \forall t \geq 0, O(F^t(x)) = O(x)\}.$$

Definition 2 (Upper Bound WU). *WU denotes the property of complete self-closure described in Axiom 4.*

Theorems

Theorem 1 (Reference Destabilizes Nullity). *If a state is referencable, it cannot remain null.*

Proof. Reference distinguishes at least two states. This contradicts observational indistinguishability required by nullity. $\square$

Theorem 2 (Unreachability of Closure). *No persistent system satisfying reference can realize WU.*

Proof. Total self-decidability contradicts standard diagonalization results for expressive self-referential systems. $\square$

Theorem 3 (Persistence Excludes Equilibrium). *No persistent trajectory can converge to a fixed observational state.*

Proof. Persistence requires infinitely many observation changes, which contradicts convergence. $\square$

Corollary 1. *Persistent systems exist strictly between WI and WU.*